Rajarata University of Sri Lanka

Department of Computing

COM 3401 – Data Communication and Networking
Laboratory 04 –Routers and Static Routing
Time: 2 Hours

Intended Learning Outcomes (ILOs)

By the end of this practical, students should be able to:

1. Introducing the concept of routing and how routers are used to connect multiple LANs
2. Getting familiar with the concept of static routing and how to configure default gateways

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Pre-Practical Preparation

1. **Reading/Resources**: Refer to the topics covered in Lecture 3 and Lecture 4

Practical Objective:

Explain how routers work in network layer and the concept of routing. Making students know how a router ca be used to connect multiple LANs in a real-world scenario. Introducing the concept of static routing and giving hands-on experience on configuring default gateways.

Practical Activity

Activity 1:

Connecting two different networks using a router

- Objectives: Connect two different LANs using router
- Instructions:
    1. Create a new project
    2. Add following devices:
        - 2 x switch (2960)
        - 4 x PC
        - 1 x Routers
    3. Design LAN1 to include two PCs connected to a switch and configure them to be in the subnet 192.168.10.0/26.

4. Design LAN2 to include two PCs connected to a switch and configure them to be in the subnet 192.168.10.64/26.
5. Connect Switch0 and switch1 with router 0 using copper straight-through cales.
6. Configure the ethernet interfaces of router.
7. Configure default gateways in every PCs.
8. Check the connectivity between LANs.

Activity 2:

Connecting multiple networks using routers and static routing

- Objectives: Connect two different LANs using routers via static routing
- Instructions:
  1. Create a new project
  2. Add following devices:
     - 2 x switch (2960)
     - 2 x PC
     - 2 x Routers
  3. Connect PC0 to the switch0 using Copper straight-through cable. Then connect switch0 to router0 using Copper straight-through cable and name it as LAN1.
  4. Connect PC1 to the switch1 using Copper straight-through cable. Then connect switch1 to router1 using Copper straight-through cable and name it as LAN2.
  5. Connect router0 to router1 using Copper straight-through cable and name the connection as WAN.
  6. Configure the LANs as follows
     - LAN1: 192.168.1.0/24
     - LAN2: 192.168.2.0/24
  7. Configure the ethernet interfaces of router
  8. Configure default gateways in PC0 and PC1
  9. Configure WAN as follows
     - WAN: 10.0.0.0/30
  10. Configure static routes in router0 and router1
  11. Check the routing table of each router
  12. Ping LAN1 from LAN0

Activity 3:

Subnetting and static routing in action

- Objective: Making students aware on how subnetting and static routing play a vital role in real-world scenario
- Instructions:
    1. Go through the given scenario

       A company has the network 192.168.20.0/24 and requires four departments.

Department	Hosts required
Administration	30
Finance	25
HR	20
IT	50

       Requirements:

    2. Calculate the network addresses
    3. Calculate the broadcast addresses
    4. Identify valid host range
    5. Build the separate topologies centred on separate switch topology
    6. Configure PCs
    7. Test communication within each LAN
    8. Connect all the switches to a router and configure the network
    9. Check inter-department connectivity